



NCCER Module 22201



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Introduction to Earthmoving





Section One

Earthmoving Operations

Objective

1. Describe earthmoving operations.
 - a. Describe the general approach to excavations and related planning.
 - b. Describe the characteristics of different types of soils.
 - c. Explain how plans and specifications are used in earthmoving.
 - d. Describe methods used to lay out slopes and grades.
 - e. Explain how to set up and coordinate operations, including setting up of staging areas, determining cycle times, and site preparation.
 - f. Describe site drainage requirements.
 - g. Explain the safety practices related to earthmoving operations.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

1. Lay out a basic earthmoving operation.



1.0.0 Earthmoving Operations

Earthmoving includes:

- Digging, loading, hauling, and dumping any earth material
- Digging and removing material from a pit or mine
- Processing the material
- Soil or rock excavation
- Backfill
- **Embankment** construction
- Control of **groundwater**



1.1.0 Excavation Planning

- The specifics of the project determine what equipment is needed.
- Each project has a set of grading plans that shows what the site is to look like after the earthmoving is complete.
- **Earth** refers to any soil or **aggregate** material in the earth's crust that is above **bedrock**.
- Different methods are required for removing soil and rock.



1.1.1 Mining Operations (1 of 2)

- MSHA establishes regulations to prevent death, disease, and injury from mining.
- **Pit** describes any open excavation of valuable materials.
- Pit operation includes heavy equipment operation, blasting, processing, and drainage.



1.1.1 Mining Operations (2 of 2)

The type of heavy equipment used for a pit operation depends on the size of the pit and the material being excavated.



1.1.2 Emergency Services

- Most jurisdictions have strict requirements for emergency vehicle access to construction sites.
- Construction sites are required to set up a unique address that allows police, fire, and other emergency vehicles to find the location easily.
- Local authorities may call for construction of special roads that allow emergency vehicles easy access to the site.
- Some jurisdictions may require standpipe or fire hydrant installations to supply water for firefighting equipment.



1.2.0 Soil Characteristics (1 of 2)

- **Gravel** — Rock pieces greater than 0.08" and less than 10" in diameter. Larger particles are called cobbles or stones, and those larger than 10" are called boulders.
- **Sand (coarse and fine)** — Mineral grains measuring 0.005" to 0.08". Sand is made from grinding or weathering rock. It usually contains a high amount of quartz.
- **Inorganic silt** — Very fine sand with particles that are 0.005" or less. Silt consists of sand that has been ground very fine. It has a tendency to absorb moisture by capillary action.
- **Clay** — Clay results from the weathering of certain types of minerals. The particles have a strong attraction to each other, so clay is very **cohesive**.



1.2.0 Soil Characteristics (2 of 2)

- **Organic matter** — Organic soils contain partly decomposed vegetable and animal material.
- **Colloids** — Colloids are very fine particles less than 40-millionths of an inch in diameter.

Soil testing and **test boring** may be done before work begins to determine what equipment is required and how to plan the work.



1.3.0 Plans and Specifications (1 of 3)

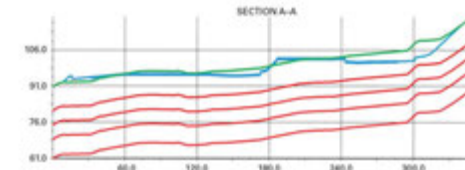
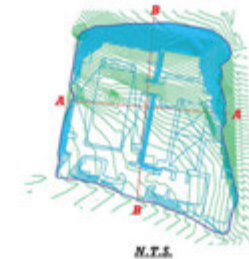
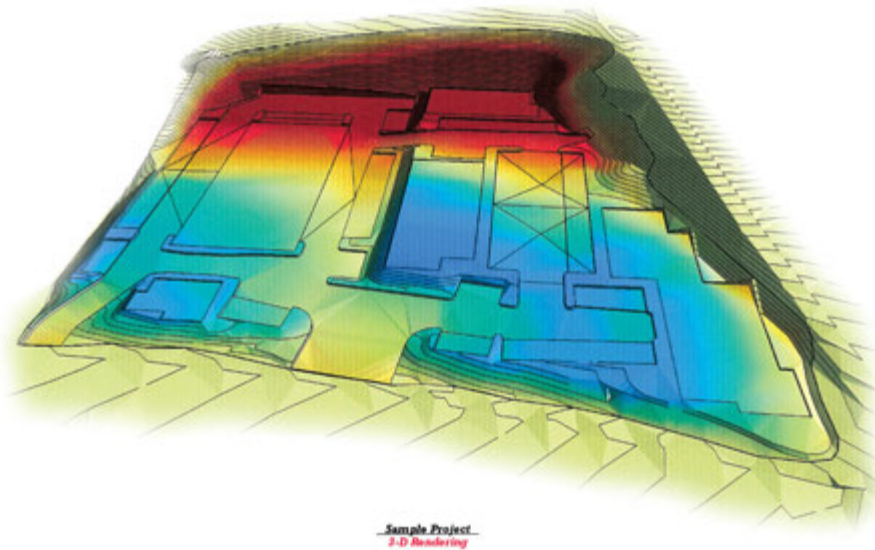
- New construction work requires a single set of plans.
- Reconstruction or rehabilitation projects have two sets of plans: the as-built plans documenting the existing roads or structures, and a set of plans for the new work.
- Project specifications provide directions, provisions, and requirements for performing the work.
- They also include standards for material specifications, test procedures, safety regulations, and traffic control devices.



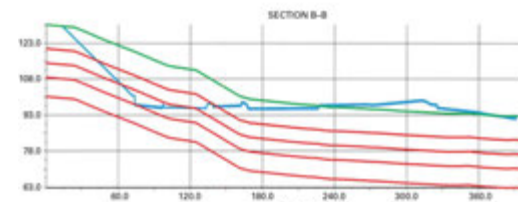
1.3.0 Plans and Specifications (2 of 3)

Left: Three-dimensional view of a job site

Right: **Cross-sections** of the same site



Scale
H: 1"=50'
V: 1"=30'



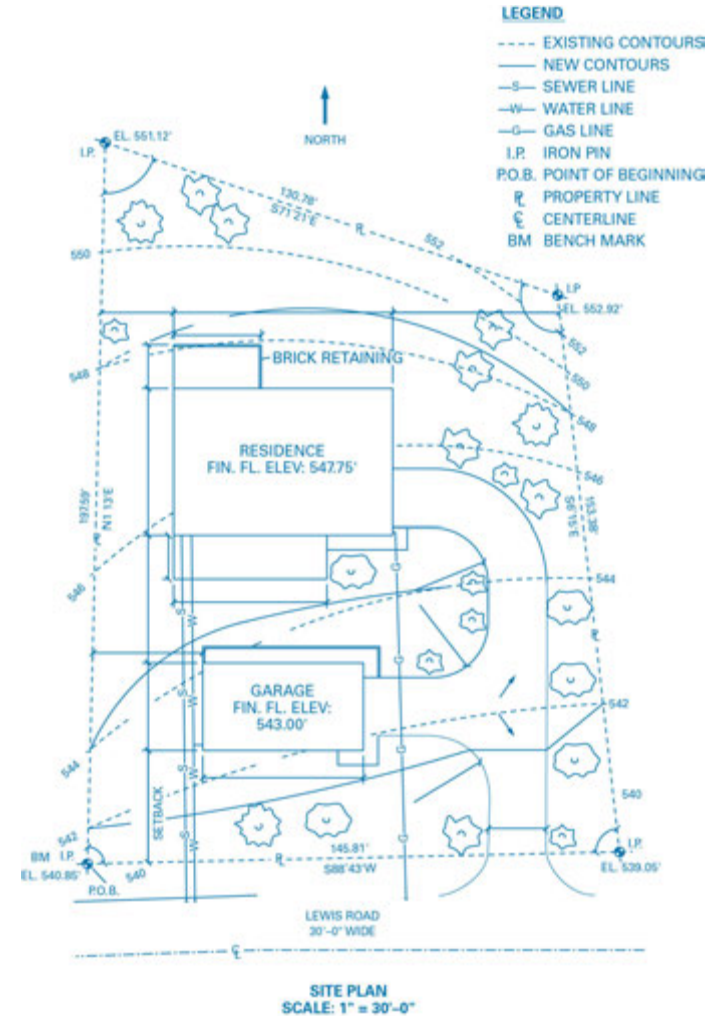
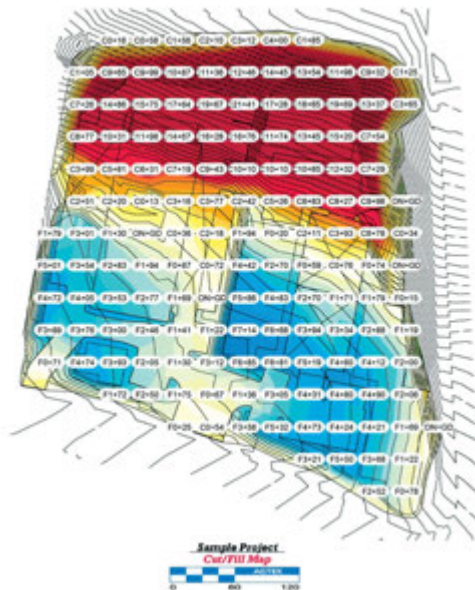
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1.3.0 Plans and Specifications (3 of 3)

Left: Cut/Fill map

Right: Site plan for part of a subdivision



1.3.1 Layout Control

- Layout controls are established by accurately setting specific **stakes** to use as permanent reference points. These are called **benchmarks**.
- Benchmarks are used for placing other stakes, such as **slope** and **grade** control stakes. All of these points are shown on site construction plans.



1.3.2 Slope Control

Slopes should be checked:

- At the completion of equipment operations but before removal from the area.
- Whenever there is doubt that the slope, stakes, and plans do not match.
- While finishing a **backslope**, **foreslope**, or **cross slope** of highway projects.



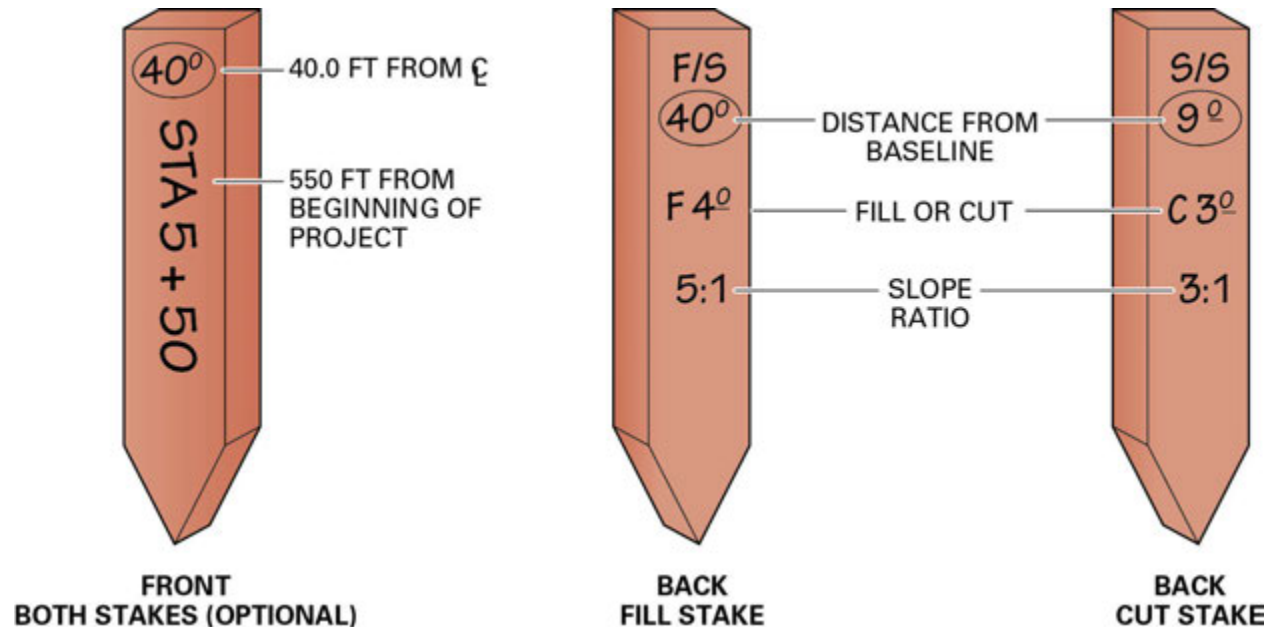
1.3.3 Grade Control

- The grade is the rise and fall of the constructed surface relative to the horizontal plane.
- **Slope stakes** provide the primary information needed to control longitudinal grades for roadway projects. These stakes normally show the vertical distance from the stakes to the subgrade shoulder points, along with cut and fill information.
- The survey crew places finish grade stakes, called blue tops, to guide finishing work and trimming of the subgrade.
- Heavy equipment operators are responsible for cutting or filling areas to the prescribed grade as stated on the grade stake.



1.4.0 Laying Out Slopes and Grades

The stakes set by the survey crew contain written grade requirements for the site. These stakes tell workers about various cut and fill requirements.



1.5.0 Setting Up and Coordinating Operations

- Site setup has different requirements depending on the type of project.
- A preconstruction conference will cover topics such as removal, disposal, emergency procedures, safety regulations, traffic control, permits, and other requirements.
- The purpose of the meeting is to bring the contractor and the owner to the same understanding of the project requirements.



1.5.1 Staging Areas, Equipment Parking Areas, and Site Access

- Site access is a concern for many contractors because of the liability that can result from accidents.
- Security of the equipment must be considered because of vandalism and theft.
- Equipment operators should know the policy and procedures for securing their equipment during non-working periods.
- Access to the site should be limited to authorized persons at all times.



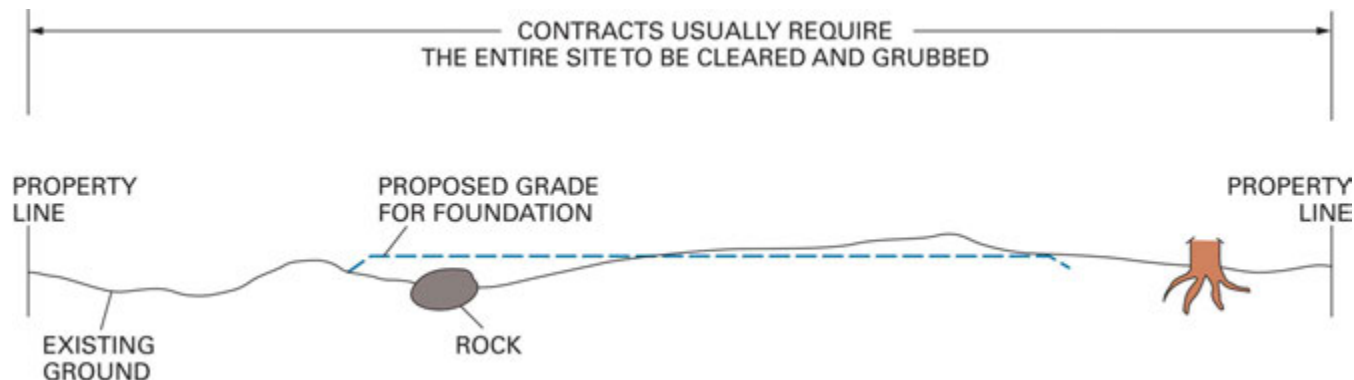
1.5.2 Construction Signs

- Post construction signs to warn the public of construction activity.
- Flaggers or other methods of traffic control may be required.
- Work may not begin until all signs and warning devices are installed according to the project plans and applicable government regulations.



1.5.3 Clearing and Grubbing

- Clearing is the removal of objects above the original ground. Grubbing is the removal of objects below ground.
- Clearing and grubbing may include the removal of structures and obstructions, including buildings, fences, old pavements, and abandoned pipe or culverts.
- All vegetation must be cleared prior to construction.
- Dispose or stockpile material as directed.



1.5.4 Maintaining Haul Roads

- On long-haul roads, a grader and water truck should make a pass over the road periodically to keep it in good condition.
- On small jobs, a grader may not be available for use on the haul road. In this case, the scraper operators can maintain the road.



1.6.0 Drainage Requirements

- Rain may impede operations because of its effect on the surface.
- Many construction sites have plans to control the flow of stormwater over the site to prevent erosion and water pollution. These plans are required by the federal government.
- Other drainage problems include control of groundwater and the allowance for drainage from embankment materials.



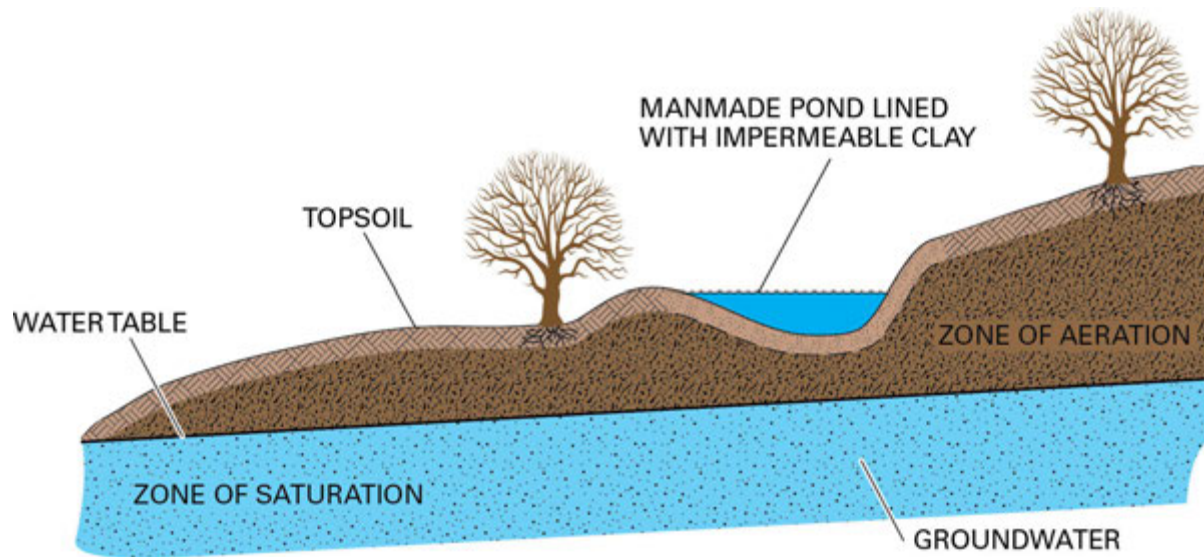
1.6.1 Control of Stormwater

- A National Pollutant Discharge Elimination System (NPDES) stormwater permit is required by the EPA.
- Contractors need to use adequate measures to prevent erosion and sedimentation and to control stormwater runoff.
- The best way to prevent erosion is to disturb only the earth that needs to be graded.
- Other erosion prevention measures include reducing the steepness of slopes, covering disturbed soil with mulch, erosion cloth, or **riprap**, and contouring the grade so that water is absorbed rather than shed.
- The best way to avoid sediment is to prevent erosion.
- Rainwater can pick up contaminants that pollute water and degrade the wildlife habitat.



1.6.2 Control of Groundwater

Soil borings are made at specific locations over the job site to plot the profile of the **water table**. From this profile, the need for any type of special **dewatering** activity, such as sumps or various types of drains, can be established.



1.6.3 Drainage of Stockpiles

- Water that drains off the stockpile must be controlled.
- When building a large stockpile of **select material**, a shallow channel should be cut around the stockpile to collect the runoff and drain it away from the storage area to a holding area.
- Silt fences must be used to slow water flow and the movement of sediment.



1.7.0 Safety Guidelines

- Safety is everyone's first priority on the job, and workers must take responsibility for their own safety.
- Once workers are familiar with the site plans, they should walk the construction area to get a firm idea of the work to be performed and be able to identify potential safety hazards.
- Report any potential problems to the supervisor.
- NEVER assume that the project engineer or supervisor is aware of a potentially unsafe condition.



1.7.1 Personal Safety

- Before starting any earthwork, walk the site to look for large depressions, large animal burrows, insect hives, boulders, stumps, or other large obstacles.
- If these issues cannot be corrected, everyone working on the job site must be alerted to their presence.
- When it is not possible to walk the site before starting work, be especially careful during the first drive over a section of land.



1.7.2 Site Contamination

While walking the site, look for signs of illegal dumping and other contamination. Workers can limit exposure to soil contaminants by wearing proper PPE.

Some warning signs of potential contamination include:

- Puddles of liquids with an unusual color or odor
- Rainbow-like reflections on surface puddles
- Unusual or foul odors
- Unmarked barrels or tanks—often buried or otherwise obscured
- Unexplained and remarkably bare patch of land in an area that is otherwise heavy with vegetation



1.7.3 Sewage

- Contact with raw sewage can place workers at risk for disease, such as hepatitis A or B.
- If illness occurs after working on a job that included exposure to sewage, tell the doctor.
- If a job requires frequent contact with sewage, talk to your supervisor about being vaccinated against this disease before exposure or symptoms appear.



1.7.4 Safeguarding Property

- Operate equipment responsibly, report defects, and secure equipment when not in use.
- Know the plans well enough to avoid damage that heavy equipment can cause.
- Never cross property lines without permission.
- Be aware of trees that need to be preserved.
- Never destroy or move survey markers or benchmarks.



Wrap Up – Trade Terms (1 of 8)

Aggregate: Manufactured or natural rock or sand that has a specific size characteristic. Used in foundations and subgrade work.

Backslope: The area between the bottom of a ditch adjacent to a roadbed and the top of the cut into the existing, natural grade.

Bedrock: The solid layer of rock under Earth's surface. Solid rock, as distinguished from boulders.

Benchmarks: Permanent points of known or assumed elevation from which surveyors can establish grades.

Center line: The line that marks the center of a roadway or other construction project. This is marked on the plans by a line and on the ground by stakes.



Wrap Up – Trade Terms (2 of 8)

Cohesive: The property of a material that sticks or clumps to itself.

Core samples: Samples of earth taken from a cylindrical test boring.

Cross section: A cutaway view of an object showing its shape and features perpendicular to its longitudinal axis or center line.

Cross slope: The pitch or grade of a road's surface perpendicular to its center line that helps drain water from the road as well as improve vehicle stability on curves.

Crystalline: Containing a mixture of closely adjoining mineral crystals. If the rock does not contain visible crystals, then it is noncrystalline.



Wrap Up – Trade Terms (3 of 8)

Cycle time: The time it takes for a piece of equipment to complete a repetitive operation. This normally would include loading, hauling, dumping, and then returning to the starting point.

Dewatering: Removing standing water from a low area using a drain or pump.

Draglines: Crane-like excavating machines configured to drop a bucket from a boom and drag it toward the machine by a cable.

Embankment: Material piled in a uniform manner so as to build up the elevation of an area. Usually, the material is placed in long narrow strips.

Expansive soil: A clay-like soil that swells with an increase in moisture content and shrinks as it dries out.

Wrap Up – Trade Terms (4 of 8)

Foreslope: The area of roadway between the shoulder/hinge point and the bottom of the adjacent ditch or natural grade.

Grade: The surface of a road, channel, or natural ground area. In civil engineering, the term usually refers to the surface level required by the plans or specifications.

Groundwater: Water that resides and moves beneath the surface of the ground.

Igneous: Identifies earth materials that were formed from formerly molten (melted) rocks

Inorganic: When considering earth materials, any substance that does not come from living organisms.



Wrap Up – Trade Terms (5 of 8)

Organic: When considering earth materials, any substance found in living organisms (plants, animals, or microorganisms) or produced by them.

Pit: An open excavation of some depth that usually does not require vertical shoring or bracing.

Riprap: Loose pieces of rock or manufactured materials placed on the slope of an embankment in order to stabilize the soil from the effects of erosional processes.

Sedimentary: Identifies rock materials that were originally deposited by wind or water.



Wrap Up – Trade Terms (6 of 8)

Select material: Earth or manufactured material that meets a predetermined specification as to some physical property such as size, shape, or hardness.

Slope: Ground that forms a natural or artificial incline upward or downward.

Slope stakes: Surveyor's stakes placed at the intersection of a cut or fill slope with the natural ground. Instructions include the slope ratio to meet the hinge point of a roadway or other reference point. Also given the terms cut or fill stakes.

Soil testing: Any mechanical or electronic test used to determine the density, moisture content, and other properties of the soil that determines the amount of compaction required.



Wrap Up – Trade Terms (7 of 8)

Stakes: Markers that are used to mark cut, fill, grade information, underground utilities, or other survey points.

Stormwater: Water directly resulting from rain or snow flowing over the ground's surface, or in natural or man-made water courses.

Test boring: A drilled or excavated hole created to take samples of the earth materials from different layers beneath the surface.

Test pit: Similar in purpose to a test boring but is a larger opening in the earth that provides a better understanding of the underlying ground structure and soil characteristics.

Trench: A long, narrow, temporary excavation that workers fill in when work is completed.

Wrap Up – Trade Terms (8 of 8)

Water table: The depth below the ground's surface at which the soil is saturated with groundwater.



1.0.0 Section Review Question 1

1. The primary earthmoving operations are excavation, backfill, and _____.
a. trenching
b. unneeded material disposal
c. embankment construction
d. grubbing



1.0.0 Section Review Question 1 Answer

1. The primary earthmoving operations are excavation, backfill, and _____.
a. trenching
b. unneeded material disposal
c. embankment construction
d. grubbing



1.0.0 Section Review Question 2

2. A very cohesive soil includes a large percentage of _____.
- a. silt
 - b. clay
 - c. gravel
 - d. organic matter



1.0.0 Section Review Question 2 Answer

2. A very cohesive soil includes a large percentage of

_____.

a. silt

b. clay

c. gravel

d. organic matter



1.0.0 Section Review Question 3

3. Labels that include the letter F and some digits distributed over a construction site plan indicate locations for _____.
a. fill application
b. cutting excavation
c. spoil areas
d. freight hauling lane width



1.0.0 Section Review Question 3 Answer

3. Labels that include the letter F and some digits distributed over a construction site plan indicate locations for _____.
a. fill application
b. cutting excavation
c. spoil areas
d. freight hauling lane width



1.0.0 Section Review Question 4

4. Stakes set by the survey crews are important because they _____.
a. identify the property owners
b. specify the kind of fill to use
c. show compass directions
d. contain written grade requirements



1.0.0 Section Review Question 4 Answer

4. Stakes set by the survey crews are important because they _____.
a. identify the property owners
b. specify the kind of fill to use
c. show compass directions
d. contain written grade requirements



1.0.0 Section Review Question 5

5. An important concern to most contractors at work sites is _____.
- a. water content of the soil
 - b. security of their equipment
 - c. the route they take to access the site
 - d. whether towns are close by



1.0.0 Section Review Question 5 Answer

5. An important concern to most contractors at work sites is _____.
- a. water content of the soil
 - b. security of their equipment**
 - c. the route they take to access the site
 - d. whether towns are close by



1.0.0 Section Review Question 6

6. Which of the following methods would NOT be very effective for controlling stormwater runoff?
- a. Reducing the steepness of a slope
 - b. Applying riprap to a slope
 - c. Cutting gullies down the face of an embankment
 - d. Contouring the grade to slow the flow of water



1.0.0 Section Review Question 6 Answer

6. Which of the following methods would NOT be very effective for controlling stormwater runoff?
- a. Reducing the steepness of a slope
 - b. Applying riprap to a slope
 - c. Cutting gullies down the face of an embankment**
 - d. Contouring the grade to slow the flow of water



1.0.0 Section Review Question 7

7. The greatest hazard to dealing with raw sewage in an excavation is _____.
- a. the stench of human waste materials
 - b. contracting hepatitis
 - c. decreased ground stability
 - d. having to compensate the owners of the damaged septic system



1.0.0 Section Review Question 7 Answer

7. The greatest hazard to dealing with raw sewage in an excavation is _____.
- a. the stench of human waste materials
 - b. contracting hepatitis**
 - c. decreased ground stability
 - d. having to compensate the owners of the damaged septic system



Next...

Section 2.0.0
Site Excavation Practices



*Read Sections 2.0.0 through 2.5.3 and
complete the Section Review.*